Project rules

Groups

max 4

students per group

- Register members and project in the **next practical class**
- At most **two groups** per project
- Groups that share a project study a different cohort, target or tissue

Preparation

1

biological question drives each project

- Start from the **question**, not from a tool
- Use **open-access data** only
- **Justify every method** you choose
- Run the **dataset check** before Checkpoint 1

Assessment

15 Dec

final evaluation: report and defence

- **Checkpoint 1:** week of 23 Nov
- **Checkpoint 2:** 4 Dec
- Both checkpoints **count toward the grade**

Research here is question-driven, not tool-driven

Every choice in the project, from the dataset to the method, traces back to one biological question.

1

The problem

State the biological or biomedical problem explicitly, and the role bioinformatics plays in solving it.

2

The data

Identify the dataset and data sources you will use, with names, accession numbers or URLs.

3

The analysis

Describe the prediction or comparison you will make, and justify the tools you selected.

Example: “Which processes separate HER2-negative breast tumours that respond to chemotherapy from those that do not?” The question picks the data (GEO GSE192341) and the method (DESeq2).

Three weeks to a final evaluation on 15 December

Weeks 1 and 2 each end with a graded checkpoint in the practical class.

1

WEEK 1

Question and data

Write the research question, access the data and run the dataset check.

Checkpoint 1
week of 16 Nov

2

WEEK 2

Run the analysis

Compare at least two methods; run the baseline and the main analysis on real data.

Checkpoint 2
week of 30 Nov

3

WEEK 3

Interpret and report

Final results and limitations; report, tables and figures; prepare the defence.

4

15 DEC

Final evaluation

Submit the report and present the defence.

Checkpoints: scores the progress regarding milestone but counts toward the grade (10% each)

Minimum outcomes at each checkpoint and the final

Bring these to the practical class: they are the minimum each group must show, not the target.

CHECKPOINT 1 · WEEK OF

16 Nov

Question and data in place

- **Research question** in one sentence
- **Dataset check** runs on the real data
- **Baseline method** and its first number
- **Roles split** and a plan B for the data

CHECKPOINT 2

30 Nov

Analysis running

- **Main analysis** run end to end
- At least **two methods** compared
- **First result** set against the baseline
- **Limitations** identified

FINAL EVALUATION

15 Dec

Report and defence

- Report **answers the question**
- **Tables and figures** support each claim
- **Limitations** and next steps stated
- **Defence** of methods and results

Checkpoints score four criteria from 0 to 2

The instructor scores every group at both checkpoints, and both scores count toward the grade.

Criterion	Level 0	Level 1	Level 2
Progress	No progress shown	Limited progress	Clear, documented progress
Technical quality	Missing or unsuitable output	Problems partly solved	Technically sound analysis
Interpretation	No interpretation	Superficial interpretation	Well-argued interpretation
Group organisation	Tasks undefined	Unclear division of work	Clear roles and responsibilities

Enzymatic degradation of polyesters

Research question: How well do sequence-based predictors of enzymatic polyester degradation generalise to unseen enzyme families and new polymer chemistries?

Dataset

- PAZy: biochemically verified plastic-active enzymes (477 unique entries), PET and PUR
- PlasticDB and PMBD for further reported enzymes and organisms; keep negative results
- UniProt for sequences; MMseqs2 to cluster enzymes into families for the held-out split
- Polymer side: repeat-unit descriptors from RDKit, optionally polyBERT embeddings

Context

Enzymatic recycling of PET already runs at demonstration scale, so companies are searching sequence databases for enzymes that attack other polyesters and polyurethanes. Only a few hundred plastic-active enzymes have been verified in the laboratory, and they cluster in a handful of families.

The problem: Measure how far a sequence-based predictor travels: hold out whole enzyme families and polymer chemistries it never saw, and report what accuracy is left.

PAZy: Buchholz et al., Proteins 90:1443 (2022), pazy.eu · polyBERT: Kuenneth & Ramprasad, Nat Commun 2023